

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/223351119>

Measurement of noise characteristics of MEMS accelerometers

Article in *Solid-State Electronics* · February 2003

DOI: 10.1016/S0038-1101(02)00220-4

CITATIONS

52

READS

1,316

3 authors:



Faisal Mohd-Yasin
Griffith University

141 PUBLICATIONS 4,309 CITATIONS

SEE PROFILE



Can Korman
George Washington University

84 PUBLICATIONS 920 CITATIONS

SEE PROFILE



David Nagel
George Washington University

191 PUBLICATIONS 3,258 CITATIONS

SEE PROFILE



PERGAMON

Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

Solid-State Electronics 47 (2003) 357–360

SOLID-STATE
ELECTRONICS

www.elsevier.com/locate/sse

Measurement of noise characteristics of MEMS accelerometers

Faisal Mohd-Yasin^{*}, Can E. Korman, David J. Nagel

Department of Electrical and Computer Engineering, The George Washington University, Washington, DC 20052, USA

Received 7 February 2002; accepted 16 April 2002

Abstract

The noise characteristics of microelectromechanical systems accelerometers at different accelerations are presented. The general experimental results show $1/f$ -type noise at low frequencies and white Gaussian noise at high frequencies. The magnitude of the noise spectral density is acceleration dependent. The results also show spectral peaks originating from the oscillators inside the accelerometers.

© 2002 Elsevier Science Ltd. All rights reserved.

Keywords: MEMS accelerometers; $1/f$ noise; White Gaussian noise

1. Introduction

Microelectromechanical systems (MEMS) are devices that have static or movable components with dimensions on the scale of a micrometer [1]. One particular device that is widely used commercially is the MEMS accelerometer. Such accelerometers typically contain some movable microbeams that measure acceleration in one or two orthogonal directions. Major markets for MEMS accelerometers are automobile airbag triggers, earthquake detection circuits and health care. MEMS accelerometers have advantages over conventional accelerometers because they are smaller, lighter and cheaper [1]. Since MEMS accelerometers are used in many systems, the noise characteristics of these devices are very important. The noise characteristics will influence the performance of the accelerometers especially when operating at lower g conditions. In this work, we report on the noise characteristics and special measurement techniques for Analog Devices ADXL202, ADXL105 and ADXL190 accelerometers.

Fig. 1 shows the functional block diagram of ADXL202 [2]. The chip contains the mechanical sensor and signal conditioning circuitry. The sensor is a surface micromachined polysilicon structure built on top of the

silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration. Deflection of the structure is measured using a differential capacitor that consists of two sets of fixed plates and central plates attached to the moving mass. The fixed plates are driven by square waves 180° out of phase that are produced by an oscillator in the chip. Acceleration will deflect the beam and unbalance the differential capacitor, resulting in an output square wave whose amplitude is proportional to acceleration. Phase sensitive demodulation techniques are used to rectify the signal and determine the direction of acceleration [2–4].

Based on the above description, one can conclude that there are three primary noise sources in a typical MEMS accelerometer measurement. The first source is from the mechanical vibration of the polysilicon springs, the second source is from the signal conditioning circuitry and the third source is from the measurement system itself. At present we are able to measure noise sources from the accelerometers at various g 's and isolate the noise source coming from the measurement circuit. We also detect a noise source originating from the oscillator inside the accelerometer signal conditioning circuitry.

2. Measurement system

Fig. 2 shows the noise measurement system. The measurement system is divided into five sections: device

^{*} Corresponding author.

E-mail address: cepus@seas.gwu.edu (F. Mohd-Yasin).

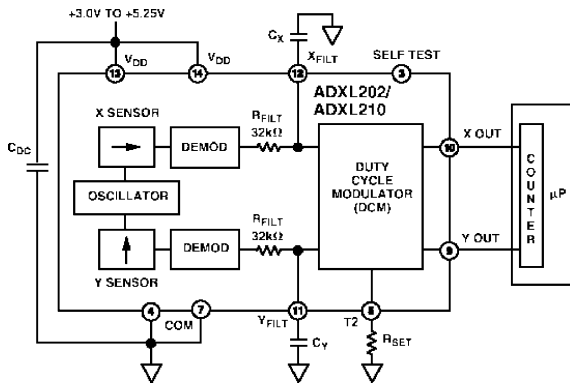


Fig. 1. ADXL202 functional block diagram.

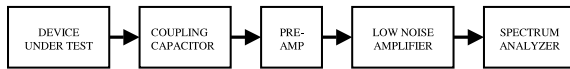


Fig. 2. Noise measurement system for MEMS accelerometers.

under test (DUT), coupling capacitor, pre-amplifier, low noise amplifier (LNA) and spectrum analyzer. Three Analog Devices accelerometers have various measurement ranges and noise characteristics. ADXL202 is an accelerometer with the measurement range of $\pm 2g$. ADXL105 is capable of measuring accelerations up to a maximum of $\pm 5g$ and ADXL190 is capable of measuring accelerations up to $\pm 100g$ [2–4]. A power supply of +4.5 V is applied to all accelerometers. The analog voltage output from the DUT is fed to the coupling capacitor. The noise signal is then fed to a pre-amplifier. In this system, one inverting amplifier is used to give a voltage gain of -36 V/V. A Linear Technology 1007 low noise operational amplifier is chosen because of its excellent low noise floor voltage of $3 \text{ nV/Hz}^{1/2}$ and its high

gain bandwidth product of 60 MHz [5]. The pre-amplified noise voltage is fed to the LNA for further amplification. An Agilent 11909A wide band LNA amplifies the noise voltage with a voltage gain of 42 V/V. After these two amplification stages, the amplified noise voltage is fed into a spectrum analyzer. The spectrum analyzer computes the power spectrum of the noise signal and displays the results in dBm. We use two spectrum analyzers in the measurement. The Hewlett-Packard 3588A is used for the low frequency measurement from 10 Hz to 150 MHz with a noise floor of -87 dBm and the Hewlett-Packard 8591E is used for the high frequency measurement from 9 kHz to 1.8 GHz with a noise floor of -74 dBm.

The measurement system utilizes several noise reduction techniques to minimize unwanted noise sources originating from outside the DUT. To minimize the effect of external electrical noise sources, all equipment uses one common ground and shielded with metal boxes. Agilent 11500A Type N cables are used for all external connections. Batteries instead of power supply from the outlet are used to supply 4.5 V to the accelerometers and +12 V/–12 V to the pre-amplifiers inside the metal shield. Placing the whole measurement system on a mechanical vibration damper minimizes external mechanical noise source. In order to minimize unwanted noise inside the measurement system, the whole circuit is laid out on a perforated board to avoid parasitic coupling. A tantalum capacitor is chosen as the coupling capacitor and metallic film resistors are used to construct the pre-amplifier circuit [5,6].

3. Measurement results

Figs. 3–5 show the noise characteristics of all three MEMS accelerometers operating at 0g. In all the figures, the total noise power spectral density (referred to as PSD in this paper) of the accelerometer being measured

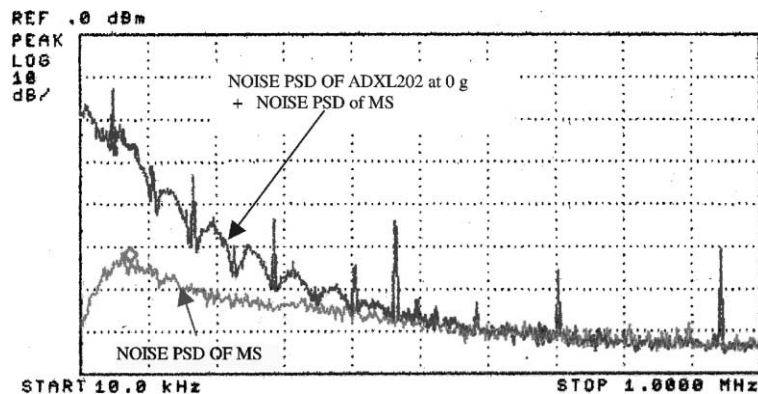


Fig. 3. ADXL202 noise PSD.

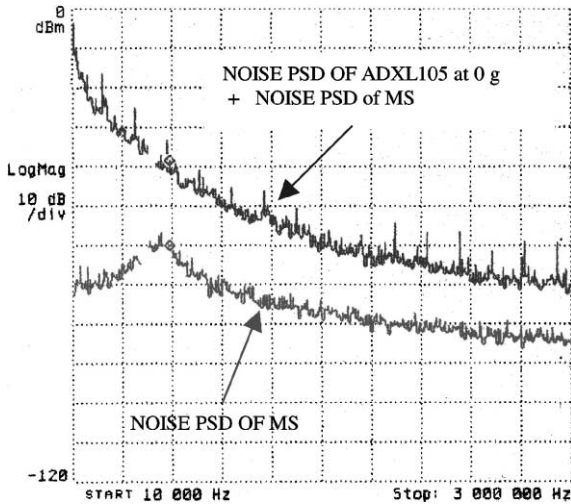


Fig. 4. ADXL105 noise PSD.

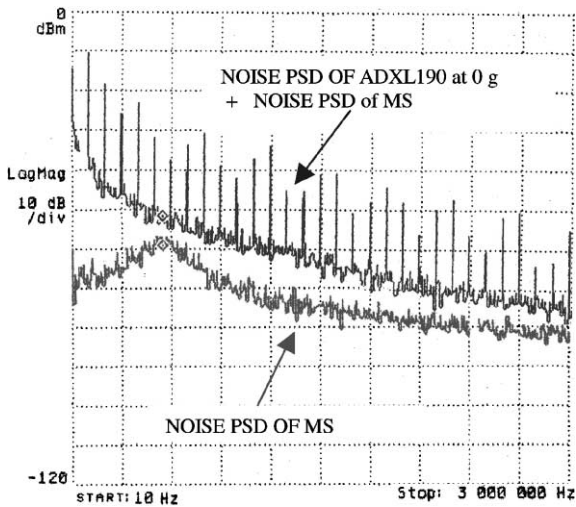


Fig. 5. ADXL190 noise PSD.

is plotted together with the noise PSD of the measurement system (referred to as MS in the figures). One can extract three important sets of information from these figures. The first is that at low frequencies the noise PSDs of all three accelerometers are higher than the noise PSD of the measurement system. This clearly allows one to extract the noise characteristics of DUT from the amplifier noise in the measurement system. The second type of information comes from the observation that the noise characteristics of all three MEMS accelerometers have $1/f$ -type noise at low frequencies at white Gaussian noise at high frequencies. The third result is the spectral peaks that can be seen in all three accelerometers noise PSDs. Fig. 6 shows a closer look of the spectral peaks of ADXL105 noise PSD. From this figure, it can clearly be seen that the peaks at points 2, 4, 6, 7 and 8 are harmonics of the oscillator operating at 200 kHz. Therefore, we can conclude that these peaks originate from the oscillator inside the accelerometer signal conditioning circuitry.

We also measured the noise PSD of all DUT's at $+1g$ and $-1g$. The noise PSDs at $+1g$ and $-1g$ have similar characteristics of noise PSD at $0g$ but with higher magnitude. One can conclude that the additional magnitude is caused by the mechanical vibration of the polysilicon springs when the DUT are at $\pm 1g$. Table 1 below shows the summary of the data that was collected from the measurement.

4. Conclusion

We have designed and built a measurement system to measure noise characteristics of MEMS accelerometers. So far we have measured the noise characteristics of three analog devices MEMS accelerometers operating at $0g$, $+1g$ and $-1g$. We conclude from the results that MEMS accelerometer noise sources have $1/f$ -type noise

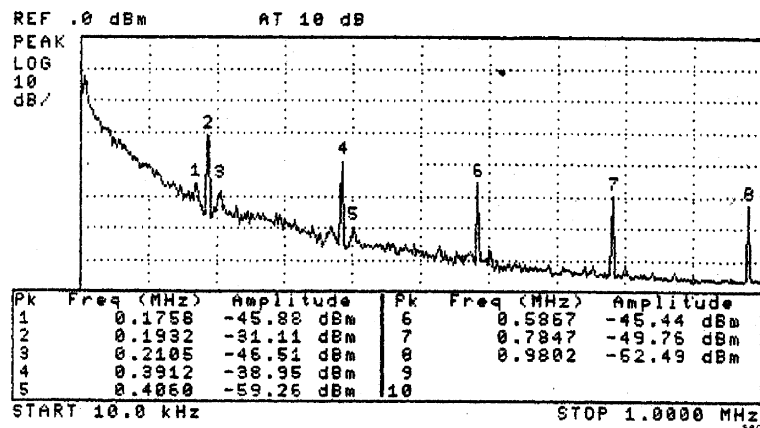


Fig. 6. Spectral peaks in the ADXL105 noise PSD.

Table 1
Summary of noise data

	ADXL202	ADXL105	ADXL190
DUT peak noise PSD ^a at 0g (dB m)	−27.02	−11.98	−38.17
DUT peak noise PSD ^a at +1g (dB m)	−24.74	−9.85	−36.88
DUT peak noise PSD ^a at −1g (dB m)	−24.31	−10.14	−37.33
Pre-amplifier peak noise PSD (dB m)	−55.45	−62.07	−61.14
Spectrum analyzer noise floor ^b (dB m)	−73.85	−86.75	−86.48
Oscillator operating frequency (kHz)	60	200	100

^a Measurement at $f = 22,510$ Hz.

^b ADXL202 and ADXL105/190 measurements use HP8591E and HP3588A spectrum analyzers, respectively.

characteristics at low frequencies and white Gaussian noise at high frequencies. The magnitude of the noise PSD at $\pm 1g$ are slightly higher than the magnitude of noise PSD at 0g. In addition, the results also show spectral peaks originating from the oscillators inside the accelerometers.

Acknowledgements

The authors would like to thank M.E. Zaghloul, J. Petrella, K. Drummond, M. Aglipay and M.Y. Afridi of The George Washington University, Harvey Weinberg of Analog Devices Inc., and M.S. Mohd of New Gen-

eration Motors for helpful discussions and other contributions.

References

- [1] Nagel DJ, Zaghloul ME. IEEE Circuit Dev Mag 1998; 17(2):14–25.
- [2] ADXL202 Datasheet, Analog Devices Inc, 1999.
- [3] ADXL105 Datasheet, Analog Devices Inc, 1999.
- [4] ADXL190 Datasheet, Analog Devices Inc, 1999.
- [5] Chang J, Abidi AA, Viswanathan CR. IEEE Trans Electron Dev 1994;41(11):1965–71.
- [6] Hung KK, Ko PK, Hu C, Cheng YC. IEEE Trans Electron Dev 1990;37(3):654–65.